The Q-value for branch D is; $Q = -931.5\ (M_{233\mathrm{Th}} + M_{\mathrm{He}} - M_{237\mathrm{U}})$. For branch A we can calculate (5.18); $Q = E_\alpha\, M_Z\, / M_{Z-2} = 4.79 \times 237/233 = -931.5\ (M_{233\mathrm{Pa}} + M_{\mathrm{He}} - M_{237\mathrm{Np}})$. By introducing values for M_{He} and $M_{237\mathrm{Np}}$ we obtain $M_{233\mathrm{Pa}} = 233.040\,108$. For branch B we calculate; $M_{233\mathrm{Th}} = M_{233\mathrm{Pa}} + 1.230/931.5 = 233.041\,428$. For branch C one obtains; $M_{237\mathrm{U}} = M_{237\mathrm{Np}} + (0.248 + 0.267)/931.5 = 237.048\,581$.

Thus all information is available for calculating branch D. The Q-value is found to be 4.23 MeV, and the $E_\alpha = 4.23 \times 233/237$ or 4.16 MeV. Although spontaneous α-decay is energetically possible, it has not been detected. The systematics of α-decay (§5.17) indicates an expected half-life of $>10^6$ y. Because the β-decay rate is much faster ($t_{½} = 6.75$ d), too few α's are emitted during the life-time of ^{237}U to be detected.

5.11. KINETICS OF SIMPLE RADIOACTIVE DECAY

Most radioactive isotopes which are found in the elements on earth must have existed for at least as long as the earth. The nonexistence in nature of elements with atomic numbers greater than 92 is explained by the fact that all the isotopes of these elements have life-times considerably shorter than the age of the earth.

Radioactive decay is a random process. Among the atoms in a sample undergoing decay it is not possible to identify which specific atom will be the next to decay. Let us make the following assumptions for a nucleus to undergo decay during a time interval Δt.

1. The probablity is independent on the history of the nucleus (i.e. equal for all t)
2. The probablity is independent on the surroundings of the nucleus
3. The probability is equal for all nuclei of a certain type and exitation energy

There are some cases where "2" is not valid but for the vast majority of cases the assumptions 1-3 are valid. Now, consider a selected nucleus and let s be the probability of decay during Δt. However, s increases proportionally to Δt according to 1. Let also p be the probability for not decaying during Δt. Then, $s + p = 1 \rightarrow p = 1 - s$ but $s = \lambda \Delta t$ and therefore $p = 1 - \lambda \Delta t$. Now, let q be the probability for not decaying during the time $t = \Sigma \Delta t$. (with $q = \Pi p$ and $p = t/\Delta t$). Since all p are equal for equal Δt one gets:

$$q = (1 - \lambda \Delta t)^{t/\Delta t} = \lim_{\Delta t \to 0} \left((1 - \lambda \Delta t)^{-1/(\lambda \Delta t)} \right)^{-\lambda t} = \mathrm{e}^{-\lambda t} \tag{5.54}$$

For a large number of atoms (>100) probability may be approximated by frequency, why (according to 3 are all q equal): $N/N_0 = \mathrm{e}^{-\lambda t} \rightarrow N = N_0 \mathrm{e}^{-\lambda t}$, or $\ln(N/N_0) = -\lambda t$

Taking the derivative with respect to t yields:

$$\mathrm{d}N/\mathrm{d}t = -\lambda N \tag{5.55}$$

where λ is the proportionality constant known as the *decay constant*.

We denote the *decay rate* by A. It is a measure of the number of disintegrations per unit time:

$$A = -\mathrm{d}N/\mathrm{d}t \tag{5.56}$$

The decay rate is proportional to the number of radioactive atoms, N, present: $A \propto N$. If 10^5 atoms show a decay rate of 5 atoms per second then 10^6 atoms show a decay rate of 50 atoms per second. If the time of observation Δt during which ΔN atoms decay is very small compared to $t_{½}$ (usually $<1\%$), one may simply write

$$A = \Delta N/\Delta t = \lambda N \tag{5.57}$$

If the number of nuclei present at some original time $t = 0$ is designated as N_0, (5.56) upon integration becomes the general equation for simple radioactive decay:

$$N = N_0\, \mathrm{e}^{-\lambda t} \tag{5.58}$$

In Figure 5.8 the ratio of the number of nuclei at any time t to the original number at time $t = 0$ (i.e. N/N_0) has been plotted on both a linear (left) and logarithmic (right) scale as a function of t. The linearity of the decay curve in the semi-logarithmic graph illustrates the exponential nature of radioactive decay. Since $A \propto N$, the equation can be rewritten as

$$A = A_0\, \mathrm{e}^{-\lambda t} \tag{5.59}$$

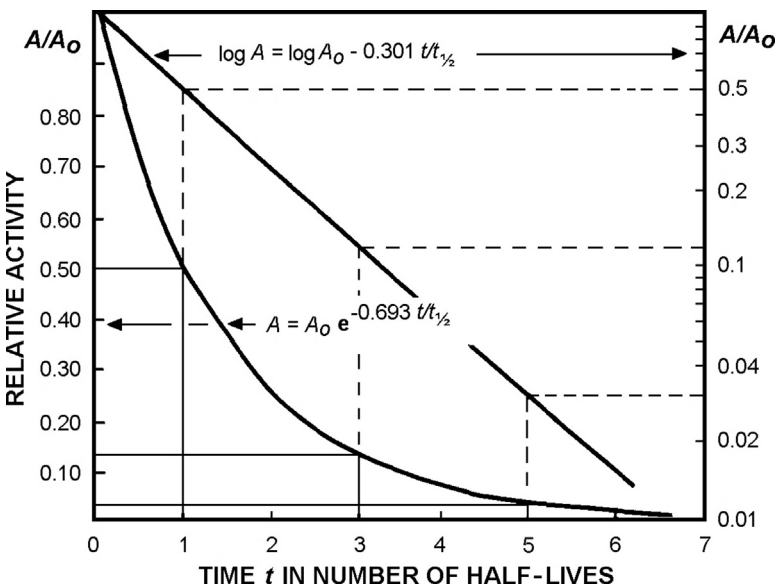


Figure 5.8 Linear and logarithm plots of simple radioactive decay.

Commonly, log A is plotted as a function of t since it is simpler to determine the disintegration rate than it is to determine the number of radioactive atoms in a sample. Instead of the decay constant λ, the average *lifetime* τ is sometimes used:

$$\tau = 1/\lambda \tag{5.60}$$

Even more common is the use of the *half-life*, $t_{½}$, which is the time needed to reduce the amount of radioactive material by a factor of 2. Thus

$$A/A_0 = N/N_0 = 1/2 = \mathrm{e}^{-\lambda t_{½}} \tag{5.61}$$

and thus

$$t_{½} = \ln(2)/\lambda = 0.693/\lambda \tag{5.62}$$

$t_{½}$ is about 70% of the average lifetime τ.

The number of radioactive nuclei remaining at any time in a sample which at $t = 0$ had N_0 atoms can be calculated from the equation

$$N = N_0/2^n \tag{5.63}$$

where n is the number of half-lives which have passed. In radioactive work, 10 half-lives ($n = 10$) is usually considered as the useful lifetime for a radioactive species since $N = N_0/2^{10} = 10^{-3}\ N_0$; i.e. N, and hence A, is 0.001 of the original N_0 and A_0.

The decay rate is usually expressed as disintegrations per second (dps) or disintegrations per minute (dpm). In measuring radioactive decay, it is very rare that every disintegration is counted. However, a proportionality exists for any particular detection system between the absolute disintegration rate A and the observed decay rate:

$$R = \psi A \tag{5.64}$$

where R is the observed decay or *count rate* and ψ the proportionality constant, known as the *counting efficiency*, cf section 9.9. This counting efficiency depends on many factors including the detector type, the geometry of the counting arrangement, and the type and energy of the radioactive decay. ψ often has a value between 0.01 and 0.5. Equation (5.64) is only valid provided $\Delta t \ll t_{½}$ (in which case $\Delta N \ll N$), where Δt is the time of measurement; this is the normal situation; cf. (5.57).

Figure 5.9 shows the radioactivity of a ^{32}P sample measured every third day with a GM counter. It is seen that the activity decreases from about 8400 cpm at $t = 0$ to 4200 cpm in 14.3 days, and to about 2100 cpm in 2×14.3 days. The uncertainty in the measurements is about the size of the circles, i.e. about $\pm$ 110 cpm at $t = 0$ and about $\pm$ 65 cpm at 30 days. In plots of this kind, the count rate measured in the absence of the sample (the *background*) must be subtracted from that obtained with the sample present to yield the correct radioactivity for the sample alone. In Figure 5.9 the

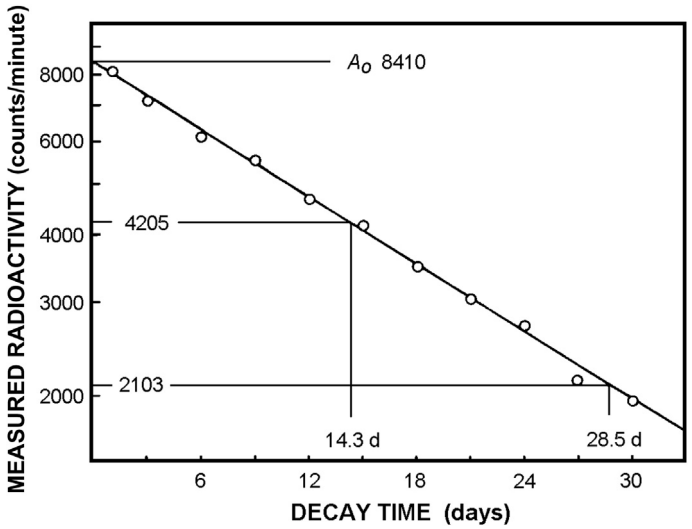


Figure 5.9 Semilogarithmic plot of the measured decay of ^{32}P using a GM counter with a background of 20 cpm.

background is so small (i.e. about 20 cpm) that it has very little influence on the decay curve.

The half-life is such a definitive characteristic of a radioactive species that knowledge of it plus the decay energy is often sufficient to allow identification of a nuclide. A radioactive sample, which exhibits a half-life of 4.5×10^9 y with α-decay energies of 4.8 MeV (77%) and 4.3 MeV (23%), is almost certainly ^{238}U as there is no other nuclide known with this exact set of properties.

With (5.57), (5.62), and (5.64) one obtains

$$R = \psi N \ln(2)/t_{½} \tag{5.65}$$

Knowing the counting efficiency ψ and the number of atoms N, the half-life can be calculated from measurement of R. For example, in a counting arrangement with $\psi = 0.515$ for α-particles, 159 cpm are observed from a ^{232}Th deposit of 1.27 mg (sample weight a). Thus $A = R/\psi = 309$ dpm, $N = a\,\mathbf{N_A}/M = 1.27 \times 10^{-3} \times 6.02 \times 10^{23} / 232.0 = 3.295 \times 10^{18}$ ^{232}Th atoms, and $t_{½} = 0.693 \times 3.295 \times 10^{18}/309 = 7.40 \times 10^{15}$ min $= 1.41 \times 10^{10}$ y.

It is quite obvious that it would not be possible to determine such a long half-life by following a decay curve like the one in Figure 5.9. Alternately, for short-lived nuclides such as ^{32}P one may use (5.65) with the known half-life and experimentally measured values of ψ and R to determine values of N.

5.12. MIXED DECAY

A radioactive sample may contain several different radioactive nuclides which are not genetically related. The decay of each nuclide follows the decay equations of the previous section. The detector measures a certain amount of the radioactivity of each species so that

$$R = R' + R'' \tag{5.66}$$

which with the introduction of (5.59) and (5.64) gives

$$R = \psi' A'_0 \, e^{-\lambda' t} + \psi'' A''_0 \, e^{-\lambda'' t} \tag{5.67}$$

Figure 5.10 shows the composite decay curve for the mixture of ^{71}Zn ($t_{½}$ 3.9 h) and ^{187}W ($t_{½}$ 23.8 h). If the half-lives of the species in the mixture differ sufficiently, as in this

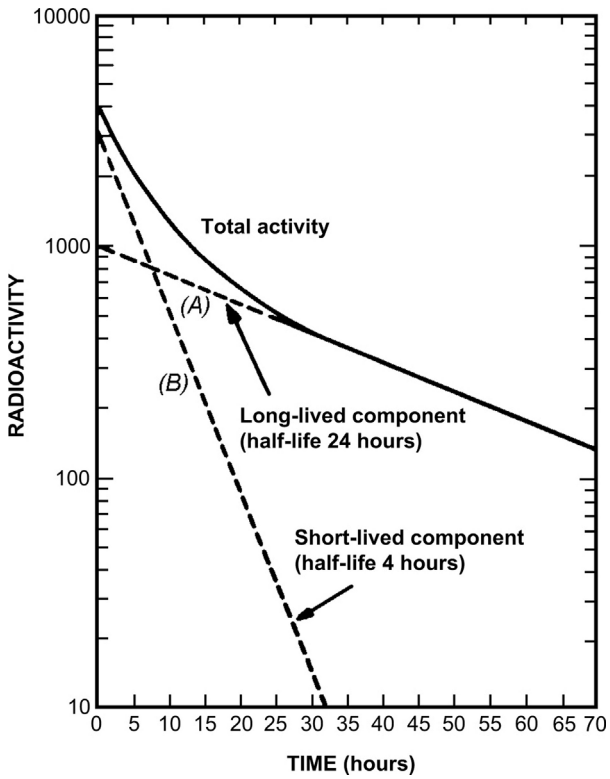


Figure 5.10 Decay diagram of a mixture of two independently decaying nuclides with half-lives of 4 and 24 h.